

Medical Robotics

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Lecture 2

Introduction to Surgical Robotics

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Contents

1	What Is a Robot?	3
1.1	Formal Definitions	3
1.2	Robotics as a Discipline: Brady’s Definition	3
1.3	What Is a Medical Robot?	3
2	Robots in Healthcare: Terminology Adopted in This Course	4
3	The Evolution of Surgery	4
3.1	Why Surgery Has Changed: The Fundamental Trade-off	4
3.2	The Spectrum of Surgical Paradigms	5
4	A Short Historical Perspective on Surgical Robotics	5
4.1	From Industrial Arms to the Operating Room	5
4.2	Timeline of Key Milestones	6
4.3	The Lindbergh Operation: A Milestone Worth Understanding	6
4.4	Market Consolidation and Its Consequences	6
5	Surgical Functions and Robot Specifications	7
5.1	The Reference Framework: From Clinical Need to Robotic Function	7
5.2	Machining of Rigid Surfaces: Orthopedic Surgery	8
5.2.1	The clinical problem	8
5.2.2	How robots help: two paradigms	8
5.2.3	Bone-mounted robots: a different paradigm	8
5.3	Constrained Manipulation: Minimally Invasive Surgery	8
5.3.1	The remote centre of motion constraint	8
5.3.2	The Da Vinci Surgical System	9
5.3.3	Active endoscope holders: the “third hand”	9
5.3.4	Competing and emerging MIS systems	10
5.4	Constrained Targeting: Interventional Radiology, Neurosurgery, and Radiotherapy	10
5.4.1	The clinical problem and the concept of constrained targeting	10
5.4.2	Key challenges	10
5.4.3	Percutaneous procedure robots	11
5.4.4	Vascular procedures and catheter robots	11
5.4.5	Open research problems in constrained targeting	12
5.4.6	Neurosurgery: historical connection and current systems	13
5.5	Micromanipulation: Microsurgery	13
5.5.1	The clinical problem and fundamental physical limitation	13
5.5.2	Robotic solutions: two paradigms	14
5.5.3	The Micron hand-held active tremor canceller	14
5.6	Surface Tracking	15
5.6.1	Transcranial magnetic stimulation robot	15
5.6.2	Tele-echography: the MELODY system	15
6	Lessons Learned	16
6.1	Costs Go Beyond the Sticker Price	16
6.2	Workflow Integration Is a System Design Problem	16
6.3	Cognitive Load and Situational Awareness	16
6.4	The Long Road to Certification	17
7	What’s Next in Surgical Robotics?	17
7.1	More Gentle Systems: Accuracy Without Aggression	17

7.2	Soft Robotics	17
7.3	Micro and Nano Robotics	17
7.4	AI-Powered Surgery	18
8	Summary: Key Conceptual Threads	19

1. What Is a Robot?

1.1. Formal Definitions

Two widely cited standards provide the accepted definitions of a robot:

RIA / ANSI definition: “A reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through variable programmed motions for the performance of a variety of tasks.”

ISO 8373 definition: “An actuated mechanism programmable in two or more axes, with a degree of autonomy, moving within its environment, to perform intended tasks.”

Both definitions share three core requirements: **programmability** (behaviour can be changed without modifying the hardware), **multi-functionality** (the same machine can perform more than one type of task), and **a degree of autonomy**. The last requirement is deliberately vague in the ISO standard — it spans the full range from a simple reactive feedback loop to a fully autonomous decision-making system. This vagueness has practical consequences: it means the boundary between a “robot” and a “powered instrument” is not always obvious, and that boundary matters enormously for regulation, liability, and clinical validation.

1.2. Robotics as a Discipline: Brady’s Definition

Michael Brady (1980): Robotics is “the intelligent connection between perception and action.”

This characterisation is arguably more useful than the mechanical definitions above, because it emphasises the *feedback control loop* rather than the hardware. In any robotic system sensors gather information about the world (**perception**), that information is processed by a controller using physical models ($F = ma$, kinematics, dynamics) and/or learned models, and actuators execute the resulting commands (**action**). The word “intelligent” implies that the mapping from perception to action is not fixed, but adaptive and context-sensitive.

In surgical robotic systems this intelligence is, most of the time, provided by a human in the control loop. Moreover, both ends of this loop are exceptionally demanding. Perception involves not just joint encoders but also force/torque sensors, tissue deformation models, and imaging modalities spanning video, ultrasound, MRI, and fluoroscopy. Action must be precise, smooth, and often ultra-gentle — operating at forces of millinewtons on structures measured in micrometres.

1.3. What Is a Medical Robot?

IEC TR 60601-4-1: A medical robot is “a robot intended to be used as medical electrical equipment or system.” More broadly, it is an active medical device that uses electrical energy.

The colloquial description from surgeon J. Moser MD — “just a Million Dollar Needle Holder” — is both a critique and a compliment. It is a **critique** because many first-generation surgical robots did exactly what a skilled human could do, only more expensively and slowly. The justification depends entirely on whether the accuracy or reproducibility gain translates into measurable patient outcome improvement. It is a **compliment** because a needle holder that never trembles, never tires, can be positioned to sub-millimetre accuracy at any hour, and integrates intraoperative imaging in a single gesture is genuinely transformative.

The key question that still drives research is: *which surgical tasks genuinely benefit from robotic assistance, and by how much?*

2. Robots in Healthcare: Terminology Adopted in This Course

Category	Scope	Primary user
Medical Robotics	Robotics for surgery, exploration, diagnosis, therapy	Clinicians / surgeons
Rehabilitation Robotics	Robots and mechatronic tools for therapy and training	Therapists
Assistive Technologies	Robots and machines that increase the personal autonomy of elderly and disabled people	Patients

Terminology note. In the broader literature these three categories are often used interchangeably. This course adopts the distinctions above consistently. When reading papers, always check which definition framework the authors are using.

The distinction between **robots** and **mechatronic tools** in the rehabilitation category is worth noting. Both share actuators and sensors, but robots carry a degree of autonomy that mechatronic tools may not. The boundary is a spectrum rather than a sharp line, and this matters for regulation: a fully autonomous rehabilitation robot would be subject to different approval requirements than a passive motorised splint.

3. The Evolution of Surgery

3.1. Why Surgery Has Changed: The Fundamental Trade-off

Surgery has always been shaped by a tension between two imperatives: **achieving therapeutic effect** and **minimising collateral damage**. Every major surgical innovation of the last century can be read as an attempt to shift this trade-off — to do more while doing less harm.

Open surgery gave surgeons direct access, rich tactile feedback, and a large working volume. The costs were severe: large incisions mean large wounds, high infection risk, long hospital stays, and significant physiologic stress to the patient. Laparoscopy, developed through the 1980s, addressed the access wound. By inserting a camera and thin instruments through small ports (typically 5–12 mm), surgeons could operate inside the abdomen with vastly reduced trauma to the abdominal wall. Recovery time for a cholecystectomy (gallbladder removal) fell from 4–6 weeks to 1–2 weeks. But laparoscopy introduced its own problems: the *fulcrum effect* at the trocar (the instrument tip moves in the direction opposite to the handle), loss of stereoscopic depth perception, limited instrument dexterity, and the need for a dedicated camera assistant.

Insight. Surgical robots did not emerge to replace open surgery. They emerged to overcome the *limitations of laparoscopy* — restoring the dexterity and vision quality of open surgery while preserving the minimal access that laparoscopy introduced. Robotic MIS is, in essence, “laparoscopy with superpowers.”

3.2. The Spectrum of Surgical Paradigms

The evolution from open $\rightarrow$ minimally invasive $\rightarrow$ endoluminal $\rightarrow$ micro/nano is not simply a trend toward smaller instruments. It is a trend toward **decoupling the therapeutic action from the access wound**. The ideal endpoint is zero external wound: deliver therapy exactly where needed, at whatever scale is required, without collateral disruption.

Paradigm	Key characteristics
Open	Direct manual access; cm–dm scale tools; forces in the N to tens-of-N range. Human hands are adequate.
Minimally invasive	5–12 mm ports; fulcrum effect; forces attenuated through trocar friction; loss of direct tactile feedback.
Endoluminal / NOTES	Access via natural orifices; flexible instruments navigating body lumens; no external incision.
Flexible robotics	Continuum robots that navigate tortuous anatomical paths rigid instruments cannot follow.
Microsurgery	Individual vessels 100–500 μm in diameter; physiological hand tremor ($\sim 50\text{--}200\ \mu\text{m}$) becomes comparable to target size.
Micro / nano	Devices on the μm –mm scale; dominant physics shift to low Reynolds number regime (viscous, electrostatic, and surface tension forces dominate over inertia and gravity).

Key concept: low Reynolds number physics. At the micro-scale, a robot cannot swim by simply waving a tail back and forth — the same motion in both directions produces zero net displacement (*Scallop Theorem*, Purcell 1977). Locomotion strategies must exploit *time-irreversible* shape-change sequences. This is why micro-robot design draws heavily on biological inspiration (flagella, cilia).

4. A Short Historical Perspective on Surgical Robotics

4.1. From Industrial Arms to the Operating Room

The first surgical robots were not designed for surgery. The PUMA 560 used in Kwoh's 1985 experiment was a standard industrial manipulator from a car assembly line. The key insight was simple but radical: if an industrial robot can place a component with sub-millimetre repeatability, perhaps it can place a brain biopsy needle with the same precision. The *surgeon-plans, robot-executes* model that this established — preoperative imaging $\rightarrow$ 3D planning $\rightarrow$ robot positions the guide $\rightarrow$ surgeon manually inserts the instrument — became the dominant paradigm for the following decade.

The PUMA was used to hold a needle guide steadily in space, exploiting the fact that a robot, unlike a human hand, does not tremble. The accuracy advantage came not only from positioning precision but also from the elimination of physiological tremor. This two-part benefit — geometric accuracy plus tremor elimination — remains the core justification for robotic assistance in any needle-based procedure.

The transition from adapted industrial robots to purpose-built surgical systems happened through the 1990s. Purpose-built systems could be designed to satisfy the specific constraints of surgery: compact enough to fit in an operating room, MR-compatible where needed, with intrinsic force limits for safety, and with interfaces designed for clinical workflows rather than programming terminals.

4.2. Timeline of Key Milestones

Year	System	Significance
1985	PUMA 560	First robot in surgery: CT-guided stereotactic brain biopsy (Kwoh et al., 22 patients). Surgeon-plans, robot-executes paradigm established.
1989	Speedy (AID)	Neurosurgery, hundreds of patients (Lavallée, Benabid et al.). First demonstration of large-scale clinical viability.
1992	ROBODOC (ISS)	First purpose-built surgical robot; total hip replacement; >70 robots, >10,000 patients. First autonomous bone-machining robot.
1994	AESOP	First robotic MIS assistant: voice-controlled endoscope holder; >400 units. Freed the surgical team from manual camera control.
1997	Da Vinci	First complete teleoperated MIS system; >850 installed systems. Master-slave architecture with motion scaling and 3D vision.
1998	ZEUS	Competing MIS system; led to the 2001 Lindbergh Operation (first transatlantic surgery: surgeon in New York, patient in Strasbourg).
2003	MARS (Mazor)	Bone-mounted robot for spine surgery. Paradigm shift: small, task-specific, intrinsically safe.
2007	NOTES (IR-CAD)	First Natural Orifice Transluminal Endoscopic Surgery with zero external incision.
2008	Da Vinci SP	Single-port access; all instruments through one umbilical incision.
2019+	Ion, Senhance, Hugo	Lung biopsy, flexible bronchoscopy, cost-effective MIS platforms. New entrants enabled by Da Vinci patent expiry.

4.3. The Lindbergh Operation: A Milestone Worth Understanding

In September 2001, Jacques Marescaux in New York performed a laparoscopic cholecystectomy on a patient in Strasbourg using the ZEUS robot, over a dedicated fibre-optic ATM link with ≈ 150 ms round-trip latency. The procedure lasted 54 minutes.

Why this matters. The bottleneck to remote surgery is not the robot — it is the **network**. A surgical robot can tolerate some latency, but a surgeon’s motor control is trained on zero-delay feedback. Latencies above 200–300 ms cause measurable performance degradation. With 5G networks and edge computing, sub-50 ms round-trip latency over public infrastructure is becoming realistic, and the clinical feasibility question is being reopened. The regulatory question — which country’s medical laws govern a surgery spanning two jurisdictions? — remains unanswered.

4.4. Market Consolidation and Its Consequences

In 2003, Intuitive Surgical acquired Computer Motion (AESOP, ZEUS), establishing a near-monopoly in robotic MIS that held until the late 2010s. This consolidation was enabled by a broad patent portfolio and by the high cost of regulatory re-approval, which made market entry by competitors prohibitively expensive.

The monopoly period had dual effects. Positively, it allowed heavy investment in system refinement, surgical training programs, and clinical evidence generation. Negatively, it kept system prices high and reduced competitive pressure to effectively solve difficult problems such as haptic feedback. The expiry of Intuitive Surgical's core patents from the mid-2010s onwards triggered the current wave of new entrants — Versius (CMR Surgical), Hugo (Medtronic), Senhance (TransEnterix) — and is expected to drive rapid price reduction and diversification.

5. Surgical Functions and Robot Specifications

5.1. The Reference Framework: From Clinical Need to Robotic Function

A central thread of this course is the methodology of associating **clinical needs** to **robotic functions**, and then identifying the robot architecture, control modality, sensors, and actuators that best address those needs. The table of surgical functions is the reference that anchors this analysis throughout the course. The five main surgical function categories are:

Domain	Robotic task	Kinematics	Control	Sensors / Actuators
Orthopedics	Machining of rigid surface	Conventional (SCARA, anthropomorphic); 5–6 DOF	Autonomous, cooperative, or hands-on	Conventional; vel: mm/s; force: up to 100 N
MIS	Constrained manipulation (RCM)	Passive mechanical or software RCM; 5–6 external + extra internal DOF	Teleoperation, shared control	Conventional; vel: cm/s; force: few N
Neurosurgery, interventional radiology, radiotherapy	Constrained targeting	Conventional + front-end with dedicated architecture; 5–6 DOF	Semi-autonomous, teleoperation, autonomous	MR/CT-compatible (pneumatic, piezo, ultrasonic); force: few N
Microsurgery	Micromanipulation	Dedicated kinematic architecture	Shared / cooperative teleoperation	Piezo, ultrasonic; force: few mN; vel: 0.7 m/s (manual)
Tele-echography, TMS, skin harvesting	Surface tracking	Conventional + dedicated wrist	Autonomous, teleoperation, shared	Conventional; vel: mm/s; force: few N

Design insight. Think of surgical robots the way we think of vehicles. A Formula 1 car and a bulldozer are both vehicles but are optimised for entirely different tasks. Designing a universal surgical robot is as misguided as designing a single vehicle to be simultaneously both. The right approach is to identify the design space for each surgical task and optimise within it.

5.2. Machining of Rigid Surfaces: Orthopedic Surgery

5.2.1. The clinical problem

Total knee replacement (TKR) and total hip replacement (THR) require removing precise amounts of bone so that prosthetic components fit together with correct mechanical alignment. If the reconstructed mechanical axis of the knee is off by even a few degrees, the prosthesis bears uneven loads and loosens prematurely — typically within 10–15 years instead of the expected 20–25.

Human surgeons use mechanical cutting guides (jigs attached to the bone) to constrain the saw blade to a planned plane. Setting the guide is itself a manual process subject to error, and soft tissue tension changes during the procedure in ways that are hard to predict.

5.2.2. How robots help: two paradigms

- **Autonomous cutting** (e.g., Tsolution One / THINK Surgical, formerly ROBODOC): the preoperative CT is segmented to build a 3D bone model, the surgeon plans the cuts on a workstation, and the robot executes them while the patient is rigidly fixtured. Positional accuracy is sub-millimetre. The trade-off is that rigid fixation and extended setup add OR time and impose patient discomfort.
- **Cooperative / hands-on** (e.g., MAKO, Stryker, formerly ACROBOT): the robot defines a *haptic virtual boundary* (a software-enforced cutting zone) and resists the surgeon's hand if it exceeds the boundary. The surgeon still drives the tool actively. This feels natural, shortens the learning curve, and has been commercially very successful. The three core features of MAKO are: enhanced preoperative planning, dynamic intraoperative joint balancing, and robotic-arm-assisted bone preparation.

5.2.3. Bone-mounted robots: a different paradigm

The Mazor Renaissance (now Mazor X Stealth Edition, Medtronic) takes a fundamentally different approach. A miniature 6-DOF parallel robot (a Stewart-platform variant) is clamped directly onto the vertebra. It positions a drill guide with sub-millimetre accuracy and the surgeon inserts the pedicle screw manually.

Key concept. Large floor-mounted robots must track patient motion using external navigation systems (infrared optical trackers), which introduces latency and complexity. A bone-mounted robot moves *with* the patient: any global motion — breathing, table shifts, intraoperative repositioning — is automatically compensated. Main advantages: minimal footprint, no patient immobilisation, intrinsically safe because the workspace is tiny. Main disadvantage: the design is anatomy-specific and workflow integration is less smooth than general-purpose systems.

5.3. Constrained Manipulation: Minimally Invasive Surgery

5.3.1. The remote centre of motion constraint

MIS instruments must pass through a trocar port in the abdominal wall and pivot about that fixed point. This is the **Remote Centre of Motion (RCM)** constraint: the instrument is a rigid body constrained to rotate about and translate through the body entry point. The effect is a reduction from 6 to 4 DOF at the tool tip (two rotations, one translation along the instrument shaft, and one rotation around the shaft), plus the kinematic inversion (fulcrum effect) that makes coordination unintuitive.

The RCM constraint can be implemented in different ways. The two main approaches are:

- **Passive mechanical RCM:** the linkage geometry constrains one point kinematically regardless of joint angles. Da Vinci uses a parallelogram linkage. Simple, reliable, requires no real-time computation.
- **Software-constrained RCM:** joint trajectories are computed in real time to maintain the pivot point. More flexible design, but requires reliable real-time computing and is less fault-tolerant.

5.3.2. The Da Vinci Surgical System

The Da Vinci system (Intuitive Surgical, Sunnyvale CA) is the world's dominant robotic MIS platform. Its three main components are the **Surgeon Console** (where the surgeon sits and operates the masters), the **Patient Cart** (the robotic arms docked to the patient), and the **Vision Cart** (image processing and display).

Key technical innovations:

- **EndoWrist instruments:** a cable-driven wrist mechanism adds 3 DOF at the instrument tip, for a total of 7 DOF at the tip. This brings full wrist dexterity inside the patient, decoupled from the RCM port constraint.
- **Stereoscopic 3D HD vision:** two parallel optical channels and a high-resolution display restore binocular depth perception lost in conventional laparoscopy.
- **Motion scaling:** the surgeon's master motions are scaled down at the instrument tip (e.g., 3:1 or 5:1), attenuating tremor and increasing effective precision. Note that force is correspondingly amplified in the reverse direction, in those version of the system that include force feedback.
- **Firefly fluorescence imaging:** the endoscope excites injected ICG (indocyanine green) dye with near-infrared light, revealing vascular and biliary anatomy through overlying tissue — activated from the surgeon console without interrupting the procedure.

Market data (2024). >10,000 Da Vinci systems installed in 72 countries. 16.94 million total procedures performed through 2024. 2.68 million procedures in 2024 alone (17% year-on-year growth; 13–16% projected for 2025). A Da Vinci procedure starts every 11.75 seconds worldwide. Average system cost: \$1.0M–\$2.3M. Primary markets: urology, gynaecology, general surgery, cardiothoracic.

On haptic feedback. Intuitive Surgical officially introduced integrated force feedback with the launch of the da Vinci 5 (dV5) system, which received FDA clearance on March 14, 2024. Previous version of the system do not provide force feedback to the surgeon. Experienced surgeons compensate by reading visual cues — watching tissue deformation rather than feeling forces. The main engineering challenges to adding haptics are: friction in the cable-driven transmission that contaminates force measurements; the long path from instrument tip to proximal sensor; and the challenge of rendering stable force feedback at the master without introducing closed-loop instability.

5.3.3. Active endoscope holders: the “third hand”

Before full master-slave systems, a simpler robotic function addressed one specific MIS limitation: holding the camera. Systems include AESOP (voice-controlled, Computer Motion), SOLOASSIST (joystick, AKTORmed Germany), FreeHand (head-motion, Prosurge UK), and research prototypes using visual servoing or gesture recognition. The key kinematic requirement for all these is compliance with the RCM constraint.

5.3.4. Competing and emerging MIS systems

Following Da Vinci patent expiry, several new systems have entered or are entering the market:

- **Versius** (CMR Surgical, UK, 2018): modular, with independent arm units configurable flexibly around the patient; reduces OR footprint.
- **Senhance** (TransEnterix / Asensus): notable for incorporating eye-tracking for instrument selection and for attempting haptic feedback — genuine technical differentiators.
- **Hugo RAS** (Medtronic): cloud-connected, targets cost-effectiveness and global scalability.
- **Ion** (Intuitive Surgical): flexible robotic bronchoscope for navigating the pulmonary tree to biopsy peripheral lung nodules — a different paradigm entirely (flexible, natural-orifice, single-use catheter).
- **Research platforms**: Raven (University of Washington) and MiroSurge (DLR Germany, with force sensors at the instrument tip) provide open platforms for algorithm development.

The **FlexDex** (US) is instructive as a non-robotic comparator: a purely mechanical, unpowered wrist attachment that adds DOF to a standard laparoscopic instrument. It addresses the same clinical problem without any electronics, enabling MIS with high dexterity for all. The drawback of this device is that it does not allow motion and force scaling, filtering, hand-eye coordination.

5.4. Constrained Targeting: Interventional Radiology, Neurosurgery, and Radiotherapy

5.4.1. The clinical problem and the concept of constrained targeting

This category covers percutaneous needle procedures (CT-guided biopsy, ablation, brachytherapy), vascular catheter navigation (coronary intervention, electrophysiology), neurosurgical instrument guidance, and stereotactic radiotherapy. The robotic task in all these cases is called **constrained targeting**: a fixed access point exists on the body surface (the skin entry point or vessel access point), and the robot must orient and advance the instrument so as to reach an internal target while respecting that constraint. The general workflow is invariant:

planning → alignment → insertion → verification

A percutaneous lung biopsy is a concrete example of this workflow. The surgical gesture consists of: positioning the needle tip at the planned point on the skin surface, orienting the needle along the pre-planned trajectory (determined by software from a preoperative CT), and then inserting the needle along that axis to reach the target — typically a pulmonary tumour node. Tissue samples are then taken for histological analysis.

5.4.2. Key challenges

- **Constrained workspace and fixed access point**: unlike laparoscopic trocars, there is no mechanical port to pivot about; the skin entry point is a soft constraint.
- **Imaging guidance and radiation exposure**: the procedure requires continuous or repeated imaging (CT, fluoroscopy) to verify needle position. Ionising radiation is dangerous for medical staff who perform multiple procedures each day. Teleoperation — allowing the operator to be physically outside the imaging suite — is therefore valuable not only for accuracy but as an occupational safety measure.
- **Soft tissue deformation and needle deflection**: biological tissue is not rigid; the needle bends and the tissue deforms during insertion, invalidating the pre-planned straight trajectory.

- **Accuracy requirements:** targets are increasingly small, requiring sub-millimetre accuracy.
- **MR/CT gantry compatibility:** operating inside an MR bore (typically 60–70 cm diameter) rules out conventional electric motors. MR-compatible actuation relies on pneumatic, piezoelectric, or ultrasonic motors, which have limited force capacity (a few newtons at most) and require different control approaches.

5.4.3. Percutaneous procedure robots

- **INNOMOTION:** one of the first MR-compatible robots (6-DOF, fully pneumatic, essentially constructed from plastic to avoid ferromagnetic interference). No longer commercialised, but influential as a design proof-of-concept.
- **Soteria:** MR-guided prostate biopsy; 5-DOF hybrid parallel-serial architecture.
- **iSYS1:** CT/fluoroscopy-guided; 2-DOF needle guide orientation with a passive positioning arm; needle insertion is performed manually by the surgeon.
- **ROBIO / Xact Robotics:** CT/PET-CT guided; 5-DOF floor-mounted; image-based registration.

5.4.4. Vascular procedures and catheter robots

Vascular procedures access internal organs through the blood vessels rather than through the skin. A catheter is introduced into a large artery (typically the femoral artery at the groin) and navigated through the vascular tree under fluoroscopic (real-time X-ray) guidance, using an imaging device known as a C-arm. A concrete example is cardiac biopsy: the catheter is advanced to the heart, where tissue samples are taken for analysis. The catheter tip is steered by cable tension that bends and flexes the distal segment, allowing it to navigate vascular curves.

The robot controls two aspects: the longitudinal advance of the catheter (telescopic insertion) and the orientation of the catheter tip (bending). The operator works from a haptic interface outside the radiation environment, viewing the fluoroscopic image on a monitor. This arrangement protects the operator from cumulative occupational radiation exposure — a concrete benefit independent of any accuracy improvement.

Animation vs. simulation — an important distinction. In the lecture, a demo video used the word “animation” for a virtual representation of a catheterisation procedure. **Animation** is pre-computed: the motion of objects is scripted and staged in advance; the control input cannot change the outcome. **Simulation** is computed in real time from a dynamic model: a physics engine integrates the equations of motion (ODEs or PDEs) at each time step, and the system state evolves in response to actual control inputs and interactions (e.g., catheter contact with vessel walls). The distinction matters in surgical robotics because simulation — not animation — is what is used for surgical training, algorithm testing, and data generation for AI. A simulator responds to what you do; an animation does not.

Representative vascular catheter systems:

- **Sensei X2 / Magellan** (Auris Health): robotic catheter systems with a 6-DOF haptic interface similar to the one shown in the lecture animation.
- **CorPath / Amigo:** simpler controller paradigms compared to Magellan/Sensei; the interface is more basic but the clinical function is analogous.
- **Niobe / Genesis** (Stereotaxis): a fundamentally different paradigm — two large permanent magnets mounted on robotic arms generate a steerable uniform field that aligns a magnetically-tipped catheter. There is no mechanical connection between the robot and

the catheter tip. Genesis achieves 70–80% faster magnet reorientation than Niobe by using lighter robotic arms with a reduced footprint. One key clinical application is the ablation of cardiac arrhythmia foci (e.g., atrial fibrillation): the catheter must maintain a controlled and reproducible contact force against the beating myocardium, a task that requires force control capabilities that are difficult for human operators but natural for a robot.

5.4.5. Open research problems in constrained targeting

Despite significant progress, a number of challenges remain active research topics:

- **Smaller, lighter, intrinsically safer robots:** most currently deployed systems are too large and have too large a footprint.
- **Prediction models for tissue deformation and needle deflection:** accurate models allow the frequency of intraoperative imaging to be reduced. An analogy from estimation theory is the Kalman filter: high-frequency but low-accuracy measurements (e.g., the robot's own sensors) are combined with low-frequency but high-accuracy measurements (imaging) to produce a continuously updated state estimate. Between images, the prediction model propagates the state forward; when a new image arrives, the estimate is corrected. The more accurate the prediction model, the smaller the correction required.
- **Real-time replanning:** needle deflection and tissue deformation mean that the initially planned trajectory is never perfectly realised. The ability to replan in real time — from an updated intraoperative model — is essential for accurate targeting.
- **Haptic feedback and stability:** force feedback in teleoperated needle insertion is critical for the operator to detect contact and tissue transitions, but it introduces a feedback loop that can become unstable if not carefully designed. This will be studied in detail in the haptics module.
- **Force sensing at the needle/catheter tip:** what the operator most needs to know is the force at the very tip — the point of contact with tissue. Placing a miniature force sensor at the tip is technically challenging (size, sterilisation, signal transmission). Estimation methods that infer tip forces from proximal measurements are an alternative.
- **Distributed force sensing along the catheter body:** during catheter navigation, the instrument can become stuck at points of vascular stenosis (e.g., calcified plaques due to atherosclerosis). The operator must be able to perceive this resistance anywhere along the catheter body, not only at the tip. Distributed sensing (fibre-optic strain sensing, for example) is an active research direction.
- **Physiologic motion compensation:** the beating heart and respiratory motion continuously displace the target. Systems must detect and compensate for this motion to maintain accurate targeting.
- **The rigidity–safety trade-off:** flexible instruments are intrinsically safer because their compliance limits the forces they can apply. However, this same compliance makes them difficult to control accurately and limits the force that can be transmitted to the tissue for therapeutic purposes. This tension between safety (requiring compliance/flexibility) and accuracy and force transmission (requiring stiffness) is a fundamental design conflict in interventional robotics and will recur throughout the course.

Key insight: enabling the impossible. CyberKnife (Accuray) illustrates the most powerful argument for surgical robots: they enable tasks that are *physically impossible* for a human. A linear accelerator is mounted on a 6-DOF anthropomorphic robot arm. The

radiation beam can be aimed from hundreds of different directions, concentrating lethal dose precisely on the tumour volume while delivering a much lower dose to the surrounding healthy tissue — a principle called stereotactic radiosurgery. Furthermore, the CyberKnife system tracks the tumour position in real time using continuous X-ray imaging, synchronising the beam direction with respiratory and cardiac motion. No surgeon can aim a beam from 200 angles at a moving target with millisecond precision. The robot makes this not only feasible but routine.

5.4.6. Neurosurgery: historical connection and current systems

Neurosurgery was the first field in which robotics entered the operating room (the PUMA 560, 1985). The rationale remains exactly the same: accurate positioning of a needle guide at a target specified in a preoperative imaging study, with accuracy that exceeds what is achievable by a human hand. Current neurosurgical guidance systems (ROSA, MedTech; PathFinder, Prosurge) follow the same paradigm — preoperative imaging → plan → registration → robot positions the guide → surgeon executes the intervention — and are now considered standard of care for stereotactic neurosurgery.

5.5. Micromanipulation: Microsurgery

5.5.1. The clinical problem and fundamental physical limitation

Microsurgery covers all surgical procedures on structures at the microscopic scale: ophthalmic surgery, some paediatric procedures, nerve repair, and microsurgical anastomosis of small vessels. The paradigmatic example is vitreoretinal surgery.

In retinal surgery, the surgeon reaches the back of the eye (the retina, a thin layer of light-sensitive tissue lining the inner back of the eye whose primary job is to capture light that has been focused by the lens, convert that light into neural signals, and send those signals to the brain via the optic nerve so you can actually “see.”) through a 0.5 mm scleral incision. Target structures — retinal blood vessels and the vessels of the retina — have diameters of 100–200 μm . One important emerging procedure is retinal vein cannulation, a procedure aiming to remove retinal vein occlusions, a leading cause of blindness. It requires the insertion of a microscopic needle (cannula) directly into the blocked vein and to inject a thrombolytic (clot-dissolving) agent. If successful, the “plumbing” is fixed, potentially providing a permanent solution with a single procedure. The challenge is that the physiological tremor of a skilled surgeon’s hand has an amplitude of approximately 50–200 μm at 8–12 Hz — comparable to the vessel diameter itself. Surgery at this scale is therefore at the physical limit of what an unaided human hand can accomplish.

There are two compounding challenges:

- **Motion scale:** the dimension of the target structures is of the same order of magnitude as the noise in the surgeon’s hand motion. Filtering or scaling this motion is necessary.
- **Force scale:** forces exchanged with retinal tissue are below the threshold of human tactile perception (approximately 7.5 mN). The surgeon cannot feel these forces directly and therefore cannot use force as a feedback signal. A robot can measure them and present them in a scaled or augmented form.

Additionally, retinal instruments enter through a single small incision, introducing the same **RCM constraint** seen in MIS: motion is restricted to rotations and translation about and along the access axis. All motion planning and control must respect this kinematic constraint.

5.5.2. Robotic solutions: two paradigms

Co-manipulation (hands-on). The surgeon holds the robot directly, and the robot's control system modifies the motion in real time. The SteadyHand robot (Johns Hopkins University) exemplifies this approach: the surgeon applies force to the instrument handle, and the robot moves cooperatively, but the tremor component is removed from the motion signal. The result is that only the low-frequency intentional motion is transmitted to the instrument tip. A key limitation of co-manipulation is that, because the surgeon is in physical contact with the instrument, motion and force cannot be independently scaled.

Master-slave teleoperation. The surgeon operates a master device and the robot moves the instrument. Because there is no direct physical contact between the surgeon's hand and the instrument, the control system can arbitrarily transform the motion signal: scale it down (e.g., 10:1 or 20:1), filter it, or impose virtual fixtures. Force can also be scaled up and rendered at the master so that sub-threshold tip forces become perceptible to the surgeon. The PRECEYES system (commercial, ophthalmic surgery) uses this approach.

Virtual fixtures. A virtual fixture is a software-defined constraint on instrument motion, analogous to a physical guide or rail but implemented in the robot controller. Hard virtual fixtures constrain the instrument exactly to a defined surface or path. Soft virtual fixtures share control between the operator's intention and the fixture: they resist motion toward the forbidden zone but yield if the operator applies sufficient force. Virtual fixtures will be studied in detail later in the course.

5.5.3. The Micron hand-held active tremor canceller

An alternative to full robotic systems is the **Micron** device (Carnegie Mellon University) — a hand-held instrument that actively cancels tremor at the tool tip without requiring a separate robot structure. It works as follows:

- Fast optical sensors measure the position and orientation of the instrument tip at high frequency.
- Three internal piezoelectric actuators deflect the tip to cancel the high-frequency tremor component.
- Cameras observe the surgical workspace and can detect anatomical structures such as vascular patterns, constructing software guides (virtual fixtures) that constrain the tip to desired paths.
- The device operates in several modes: tremor compensation only (shelving filter that removes frequencies above the intentional motion band); soft virtual fixtures (shared control between operator and fixture); hard virtual fixtures (exact constraint to a defined path or surface).

This device illustrates an important design principle: a robotic solution is not always the only answer. A carefully designed mechatronic instrument, with sophisticated sensing and control, can achieve similar functional goals with a much simpler system architecture that integrates naturally into existing surgical workflows.

Force perception threshold. At forces below ≈ 7.5 mN, surgeons cannot feel what they are doing — yet this is the relevant force range for peeling epiretinal membranes. Sensorised instruments with strain gauges can measure these sub-threshold forces and present them visually. Studies show that even visual (not haptic) force display significantly reduces intraoperative tissue damage.

5.6. Surface Tracking

Surface tracking is required when a sensor or therapeutic device must maintain controlled, reproducible contact with a body surface — as in tele-echography (ultrasound), transcranial magnetic stimulation (TMS), and surface-based radiotherapy. The core robotic challenge is regulating contact force, while compensating for patient motion (breathing, involuntary movement).

5.6.1. Transcranial magnetic stimulation robot

TMS is used to treat neurological and psychiatric conditions including epilepsy, depression, and chronic pain. A strong magnetic coil placed against the skull induces localised electrical currents in the cortex. The position and orientation of the coil determines which cortical region is stimulated; precision and session-to-session reproducibility are therefore critical. A clinical TMS session may last 20–40 minutes, during which a human operator must hold the coil steadily against the patient’s head — an ergonomically demanding task that also reduces reproducibility.

The TMS robot (Axilum Robotics, France) uses a UR10 collaborative robot (this robot is also present in the Sapienza robotics lab). A magnet is mounted at the robot’s end-effector and placed on the skull. The robot maintains the prescribed coil pose throughout the session using force control and real-time optical head-tracking. Contact force is regulated to below 10 N — roughly equivalent to pressing with the weight of a 1 litre bottle of water. This is sufficient to maintain contact without causing discomfort. Because the robot tracks the patient’s head, involuntary motion is automatically compensated.

An alternative approach uses a passive articulated arm with optical markers on both the coil and the patient’s head to compute and display the positioning error in real time, allowing a human operator to correct manually. This does not provide the reproducibility of a robot, but is a simpler and more affordable intermediate solution.

5.6.2. Tele-echography: the MELODY system

Ultrasound examination (echography) requires a trained specialist to position and orient the probe, adjust contact pressure, and interpret the resulting image. In regions with limited access to specialist sonographers, a teleoperated echography robot allows a remote expert to conduct the examination over a network connection.

The MELODY system (AdEchoTech, France) places a robotic manipulator at the patient site. The remote specialist holds a simple hand-held controller — essentially a rigid object instrumented with an IMU (inertial measurement unit) and a force sensor — and their hand motions are transmitted to the robot, which moves the ultrasound probe accordingly.

Why a simple rigid object suffices as the master. The operator must transmit position, orientation, and contact force to the probe. An IMU provides 6-DOF orientation and, after initialisation, relative position. A calibrated spring (or a commercial force sensor) registers applied pressure. No complex haptic display is needed for the basic functionality: the operator is simply issuing motion commands, not receiving force feedback. The simplicity of the master device is a deliberate design choice — it reduces cost and makes the system deployable in low-resource settings.

The robot must regulate contact force (impedance control) to ensure the probe maintains adequate tissue coupling without causing patient discomfort or moving the probe too deep. As the probe moves over the body surface, soft tissue deforms and the acoustic window changes with depth. The course will return to the control methods for this type of interaction in the force control module.

6. Lessons Learned

After more than 30 years of medical robotics research and clinical deployment, a clear picture has emerged of the factors that determine whether a system moves from the laboratory to clinical adoption. The ratio of research prototypes developed to systems actually adopted in clinical practice is very small: many technically successful systems have never reached the standard of care.

6.1. Costs Go Beyond the Sticker Price

The total cost of ownership of a robotic surgical system includes: capital cost (\$1–2M for Da Vinci); annual service contract (typically 10–15% of capital cost); per-procedure disposable instrument cost (each EndoWrist has a limited use count before mandatory replacement); additional OR time particularly during the learning curve; and staff training. Alongside this, the adoption of a robotic system often requires modification of the operating room itself — changes to layout, infrastructure, and workflow. A robot that requires the OR to be redesigned is unlikely to be commercially adopted, regardless of its technical performance.

These must be weighed against: reduced hospital length of stay, fewer complications, and access to new procedure types. Health technology assessment bodies (NICE in the UK, HAS in France, AIFA in Italy) apply formal cost-effectiveness analysis — cost per QALY (quality-adjusted life year) — to determine whether coverage is justified. The verdict is sensitive to the specific procedure and health system context: higher bed costs favour robotic adoption; lower bed costs do not.

6.2. Workflow Integration Is a System Design Problem

A robot that disrupts surgical workflow creates risk. The OR is a precisely choreographed environment where staff have defined roles, equipment is arranged to specific protocols, and communication follows established patterns. One of the primary design concerns in developing a new medical robot must be easy integration into existing workflows and existing operating rooms. This requirement is in direct tension with the engineering drive toward more capable, more complex systems.

The Da Vinci patient cart requires a trained bedside assistant who can exchange instruments, retract tissue, apply suction, and take over manually in an emergency. The cost and complexity of training this assistant is part of the implementation burden.

6.3. Cognitive Load and Situational Awareness

Robotic systems can paradoxically *increase* the cognitive load on the surgeon rather than reducing it, at least during the learning period. Operating from a console, the surgeon's visual attention is captured by the endoscopic display. The broader OR — vital sign monitors, the anaesthesiologist, the bedside assistant — is outside direct visual field.

Situation awareness. Situation awareness is the operator's real-time understanding of what is happening, why it is happening, and what is likely to happen next. It is a concept from cognitive science and human factors engineering that is directly applicable to surgical robotics. An operator who has lost situation awareness — because too many things are happening simultaneously, or because the interface is too complex, or because critical information is not presented clearly — is at high risk of error. Designing robotic systems that *support* and *extend* rather than *reduce* the operator's situation awareness is one of the central human factors challenges in surgical robotics.

Engineers and researchers must navigate the tension between system capability (which tends to increase complexity) and operator situation awareness (which benefits from simplicity and clarity). This is not a technical problem that can be solved by adding more sensors or better algorithms; it is a human-centred design problem.

6.4. The Long Road to Certification

Regulatory approval for a new surgical robot takes a minimum of 7–10 years in the EU and USA, and sometimes considerably longer before a system achieves widespread clinical use. The process requires extensive preclinical testing, first-in-human studies, multi-centre randomised trials, and post-market surveillance. No regulatory framework specifically written for surgical robots exists; applicable frameworks are composites of industrial robot standards, medical device regulations, and software device guidance. This timeline is a structural barrier to innovation: by the time a system completes certification, the technology may have evolved substantially.

7. What's Next in Surgical Robotics?

7.1. More Gentle Systems: Accuracy Without Aggression

A general trend in surgical robotics is toward systems that achieve the required accuracy while being *gentler* — exerting smaller forces on tissue, recovering more quickly from perturbations, and adapting to patient variability. Soft robotics, as discussed above, is one pathway. Another is context-aware control: for example, exoskeletons used in rehabilitation do not need to provide constant assistance. An ideal exoskeleton would detect when the user is fatiguing (e.g., by monitoring heart rate and electromyographic activity) and increase assistance only when needed, synchronising with the user's own effort. This requires wearable devices that communicate with each other and with a central control system — an architecture that is becoming feasible with current sensor and wireless communication technology.

7.2. Soft Robotics

Instruments made from elastomers, hydrogels, and compliant polymers are intrinsically safer near human tissue: compliance limits the forces they can apply even in the event of a control failure. Soft robots can also navigate tortuous anatomical paths — the Ion flexible bronchoscope is a commercial example. Engineering challenges include: sensing on compliant substrates, pneumatic tether requirements, nonlinear and history-dependent mechanics, and control design without standard rigid-body kinematic models.

7.3. Micro and Nano Robotics

Miniaturised devices operating inside the body at scales inaccessible to current instruments. Biomedical applications include targeted drug delivery, brachytherapy, localised hyperthermia, biopsy, ablation, stenting, and scaffolding for tissue engineering.

Working at the micro and nanoscale completely changes the engineering picture. The challenges are not merely miniaturisation of existing approaches: the dominant physics change, power delivery becomes difficult, locomotion in biological fluids requires entirely new strategies, and localisation of a device deep inside the body is a non-trivial inverse problem.

Key research challenges:

- **Locomotion in biological fluids:** micro-devices must move through blood or intestinal contents. Blood is a viscous fluid, and at microscopic scales the Reynolds number is very low, meaning viscous forces dominate over inertia. Conventional propulsion strategies do

not work; biological designs (flagella, cilia) provide inspiration. Pills that can be ingested to examine the intestine already exist, but they rely on passive peristaltic transport — they cannot be steered to a specific location.

- **Magnetic actuation:** the leading active actuation approach uses external variable magnetic fields to drive magnetically responsive structures. Hybrid bio-magnetic robots have been demonstrated: sperm cells coated with magnetic nanoparticles whose direction of locomotion is steered by an external field. At the milli-scale, legged robots with ferromagnetic limbs can be made to walk by cycling the external field. Controlling the locomotion pattern of these devices and, more importantly, controlling multiple devices independently, remains an open research problem.
- **Localisation via Electrical Impedance Tomography (EIT):** EIT injects small electrical currents through surface electrode arrays and measures the resulting voltage distribution to reconstruct a 3D image of the internal electrical impedance. A micro-robot with different impedance from surrounding tissue appears as a detectable anomaly. The reconstruction is an inverse problem described by partial differential equations, and solving it in real time is computationally demanding. Neural network approaches are being developed to achieve real-time localisation from EIT measurements.
- **Manipulation via non-uniform magnetic fields:** by shaping spatial field gradients, non-contact forces and torques can be applied to micro-particles, allowing assembly and manipulation of micro-scale structures. Needle-shaped electromagnets embedded in a macro robot provide localised, high-gradient fields for fine manipulation.
- **Macro/micro integration — the REGO project:** because localisation and control of free-floating micro-robots over large distances inside the body is extremely difficult, one strategy is to bring the micro-robots close to the target using a macro carrier robot first. The REGO project explores this approach: a macro manipulator navigates to the target site and deploys a swarm of milli-scale robots, which are then controlled locally by magnetic fields from miniature electromagnets in the carrier.

Interdisciplinary scope. Micro-robotics research requires soft matter physics, colloidal chemistry, microfabrication, control theory, MRI physics, cell biology, and pharmacology — simultaneously. This breadth is why progress has been slower than in MIS robotics and why the potential impact is correspondingly greater. Targeted cellular-level therapy would represent a paradigm shift comparable to the discovery of antibiotics. This is blue-sky research: no micro-robot has yet been deployed in clinical practice, but the research foundations are being built now.

7.4. AI-Powered Surgery

Current surgical robots are tools: they execute surgeon commands with enhanced precision. AI is progressively adding layers of autonomy. A useful taxonomy (Sheridan 1992, adapted) describes levels from “robot offers no assistance” to “robot performs the entire procedure independently.” Present commercial systems sit at levels 1–3. Research is exploring levels 4–6.

Near-term applications:

- **Computer vision:** real-time instrument tracking and anatomical structure recognition in the endoscopic image; context-aware assistance.
- **Skill assessment:** kinematic and video-based models trained on expert data that evaluate trainee performance and identify error-prone motion patterns.

- **Autonomous subtask execution:** automated performance of well-defined, bounded sub-tasks (cutting along a marked line; specific suture knot) under surgeon supervision.
- **Decision support:** alerts for anatomical danger zones based on preoperative image registration; flagging of anomalous force patterns; optimal port placement suggestions.

AI and simulation. Simulation is becoming increasingly important in surgical robotics for two distinct reasons: (1) training surgeons in a safe environment without patient risk, and (2) generating large labelled datasets for training AI algorithms, which require far more data than clinical cases alone can provide. The quality of the simulation — in particular, the realism of tissue mechanics — directly determines the value of the generated data.

Liability and autonomy. If an autonomous surgical robot makes a decision that leads to patient harm, who is responsible: the surgeon (the legal operator), the hospital, the manufacturer, or the AI system? Current legal frameworks do not answer this. This uncertainty is a genuine and independent barrier to clinical deployment of high-autonomy systems, distinct from any technical feasibility question.

8. Summary: Key Conceptual Threads

- Surgery's fundamental trade-off is **therapeutic effect vs. collateral damage**. Every generation of surgical technology has pushed this frontier. Robots are the latest, not the last, instrument in this progression.
- **Different surgical tasks require completely different robots.** The design space for each is defined by the physics of the tissue interaction, the required accuracy, the force range, the imaging modality, and the clinical workflow.
- **Clinical outcomes, not engineering metrics, determine value.** A technically impressive robot that does not improve patient outcomes — or is not economically justifiable — will not succeed clinically. The ratio of research prototypes to clinically adopted systems is very small.
- **Easy integration into existing workflows is a primary design constraint.** A system that requires the OR to be redesigned will not be adopted.
- **Autonomy is a spectrum.** Surgical robots will move up that spectrum gradually, constrained by the safety and regulatory requirements that the clinical setting demands.
- **Simulation, not animation:** simulation computes system evolution in real time from dynamic models and responds to actual control inputs. Animation is pre-staged. Surgical robotics needs simulation — for training, for planning, and for AI data generation.
- **The frontier is at the intersection of disciplines.** The researchers who will make the next major contributions will be those who can bridge engineering, biology, clinical practice, and ethics simultaneously.

Key topics for revision.

1. Why different surgical domains require different robots.
2. The specific advantages of the Da Vinci surgical system.
3. What makes micro-robotics physically different from macro-robotics.
4. The distinction between animation and simulation, and why simulation matters for surgical robotics.

5. Virtual fixtures: definition, hard vs. soft, and the two paradigms for their application in microsurgery.
6. The rigidity–safety trade-off in flexible instruments and how it manifests in constrained targeting.
7. Lessons learned from clinical deployment.